Clearer Waters: an Analysis of Marine Debris Detection Models

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***Abstract*: Marine debris detection from underwater ROV imagery is a challenging task, made difficult by colour cast, low contrast, small objects and heavy class imbalance. Using the TrashCan-Material (7,212 images, 16 classes), we benchmark COCO-pretrained Faster R-CNN baselines with MobileNetV3-Large-FPN and ResNet-50-FPN backbones, then introduce improvements aimed at the dataset's known challenges: repeat-factor sampling (RFS) for rare classes, underwater colour and contrast augmentation, and an upgraded ResNet-50-FPN v2 detector. We also compare against a single-stage RetinaNet trained with focal loss. Our improved Faster R-CNN raises Test AP50 from 18.9% to 21.4% (Test AP from 10.7% to 12.1%) at 49 FPS, and outperforms RetinaNet (11.6% Test AP50), suggesting two-stage detectors are better suited to sparse underwater scenes. Small objects and the catch-all trash_etc class remain the main failure points.**

Keywords: marine debris detection, object detection, Faster R-CNN, RetinaNet, class imbalance, underwater imagery, TrashCan

I. INTRODUCTION

Vision equipped autonomous underwater vehicles (AUVs) can be deployed to detect and remove underwater trash, to address the urgent and damaging impact of pollution on marine ecosystems. However, precise trash detection using these vehicles requires the development of highly trained neural networks. Through this paper, we aim to benchmark state-of-the-art (SOTA) models using the framework presented by [2] on the dataset presented in [1]. We also aim to implement improvements and variations of the benchmarked models, to address known challenges in trash detection and segmentation in real-world and noisy marine environments.

II. RELATED WORK

Effective trash identification and clean-up using AUVs requires that they are equipped with precise and SOTA vision-based trash detection models. Known neural scaling laws demand large annotated datasets be used for training and validation, to scale and improve model performance. J-EDI (JAMSTEC E-Library of Deep-sea Images) is one such dataset, containing videos from ROVs, as operated by the Japan Agency of Marine Earth Science and Technology. Only a small portion of the videos contain marine debris. Still, using those videos, [1] presents TrashCAN: a dataset of underwater trash, sourced from J-EDI and annotated using bounding boxes and segmentation masks. The dataset contains 7,212 images classed as either trash, rov, bio or unknown. Trash images are further tagged per material and instance, creating two versions of the dataset. Alongside the release of this new dataset, [1] shares results of baseline detection and segmentation experiments, evaluated using COCO AP metrics. A preliminary finding in these experiments was that models trained on instance versions of TrashCan scored marginally higher than the material version, instance-based classes created a more visually similar grouping than material based classes. For detection experiments (object detection), Faster R-CNN with a ResNeXt-101-FPN architecture was trained and evaluated (pre-trained over COCO dataset for 10,000 iterations). For segmentation experiments, a Mask R-CNN with X-101-FPN architecture was chosen (also pre-trained with COCO for 15000 iterations).

Following the work of [1], TrashCAN was used together with DeepTrash (another dataset capturing the more varied, harsh and real-world conditions of marine environments), by [2] to address the critical lack of benchmarking frameworks in marine debris detection and comparative analysis of model/architecture quantitative performances. [2] identifies strengths and weaknesses of models in handling marine object detecting by analysing the smoothness of their training/loss curve alongside metrics like Precision, Recall, F1 score and mAP. Evaluated models include: Faster RCNN, SSD, YOLOv8 and the more improved YOLOv9. In their comparative analysis, YOLOv9 was found to stand out in its exceptional training stability (smooth loss function), speed and accuracy (achieving the highest precision, recall and mAP scores). Inversely, YOLOv8, Faster R-CNN and SSD displayed lower stability with a fluctuating loss function, with SSD demonstrating the lowest stability. While YOLOv9 dominates overall (credited to its PGI and GELAN features), authors of [2] note that challenges remain in accurately detecting uncommon objects, e.g. “shipwrecks” in noisy environments.

Unlike the optical datasets proposed by [1] and used in [2], [3] classifies marine debris using Forward Looking Sonar (FLS) images, which are greyscale and commonly used for underwater imagery. The specific dataset used was the Marine Debris FLS dataset, which contains 1,865 FLS images annotated with bounding boxes and classed as either bottle, can, chain, drink can, hook, propeller, shampoo bottle, standing bottle, tire or valve. These images were not taken in a real-world and open water marine environment, they were instead taken underwater of an Ocean Systems Lab water tank at Heriot Watt University, using an ARIS Explorer 3000. Most prior methods were trained on small datasets and were class agnostic, [3] proposes a more class dependent system for classifying debris. Images are preprocessed using histogram equalisation and a median filter (to improve contrast and reduce noise), then augmented through flips and rotations, tripling the training set size. For detection, a Faster R-CNN was trained with both VGG16 and ResNet-50 backbones (pretrained over ImageNet). Faster R-CNN with ResNet-50 performed best. It’s worth noting however that this paper reports results should be treated with caution, as just one baseline comparison was reported, and data used for training is not representative of an open marine environment.

III. METHODS

1. Baseline Detectors

For our baselines we chose the two-stage Faster R-CNN [4] with a Feature Pyramid Network (FPN) [5], as it is the detector used by [1] and so gives the most direct comparison against the TrashCan benchmark. Two backbones were trained, both pretrained on COCO [10]: a MobileNetV3-Large-FPN [9] (19.0M parameters), chosen as a lightweight, high-FPS option, compatible for onboard AUV use, and a ResNet-50-FPN v1 (41.4M parameters) as a stronger and more accurate reference. In both cases the COCO box predictor was replaced with a 17-way RoIAlign head (16 classes plus background) and the full model was fine-tuned on TrashCan.

2. Improvements

Our exploratory analysis revealed two challenges: a long-tailed class distribution and degraded underwater colour and contrast. We suggest the following improvements to address these challenges:

- Class-weighted RoI loss with weight ensembling: Inverse square-root class-frequency weights were applied to the RoI classification loss to lift tail classes. To avoid losing accuracy on head classes, the fine-tuned and pretrained weights were then interpolated in weight space using WiSE-FT [6] ($\alpha = 0.50$).
- Repeat-factor sampling and underwater augmentation: RFS [7] oversamples images containing rare classes (image frequency below 6%), so tail classes are seen more often without changing the loss. This was paired with red-channel colour-cast compensation and contrast/brightness jitter, motivated by the weak red channel and low contrast found in a large share of frames. Combined with an upgraded ResNet-50-FPN v2 detector, this forms our final improved model.
- Single-stage alternative: To test whether a dense single-stage detector handles underwater imagery better, we trained a RetinaNet with a ResNet-50-FPN v2 backbone and focal loss [8] ($\gamma = 2.0, \alpha = 0.25$), which down-weights easy background anchors.

IV. EXPERIMENTS

1. Dataset

We use TrashCan 1.0 [1], built from the J-EDI (JAMSTEC E-Library of Deep-sea Images) archive of ROV videos, mostly recorded in the Sea of Japan, with frames extracted at one per second. It contains 7,212 images annotated in COCO format with bounding boxes and instance masks, and is available from the University of Minnesota Conservancy (doi:10.13020/g1gx-y834). Of its two label versions, all experiments here use the Material version: 16 classes and 12,336 boxes, with trash split by material (e.g. trash_plastic, trash_metal) alongside rov, plant and animal classes.

3. Exploratory Data Analysis

Class imbalance is severe. The rov class alone holds 3,317 boxes (27%), 23 times more than trash_rubber (142), and the top three classes hold 59% of all objects. Trash makes up 50% of objects and appears in 72% of images, while trash_etc acts as a catch-all mixing many object types. Scenes are sparse, with a mean of 1.7 objects per image and 58% of images containing a single object. Of trash objects, 18% are small, 50% medium and 32% large by COCO size buckets, the median box covers 4.1% of the image, and 32% of boxes are elongated beyond a 2:1 aspect ratio.

Image quality is also a challenge: 42% of images have a weak red channel (median red ratio 0.82), 23% are low contrast and 15% are dark. Frames come in two resolutions (480 by 270 and 480 by 360), some with black bars and burned-in telemetry text. Importantly, frames are highly correlated: the 7,212 images come from just 312 videos (median 17 frames each). A random frame-level split would therefore leak near-identical frames between training and testing, so we use video-disjoint splits of 4,989 train, 649 validation and 1,574 test images, with no video shared across splits.

4. Training and Evaluation

The Week 7 baselines were trained with AdamW and uniform sampling, and both plateaued early (Val AP50 of 20.2% for MobileNetV3 and 22.7% for ResNet-50 v1), over-fitting to background RoIs. The improved recipe instead uses SGD (momentum 0.9) with linear warmup and a cosine learning rate schedule, freezes early backbone stages 1 to 3, and adds RFS and underwater colour/contrast jitter.

All models are evaluated with the exact COCO protocol: 101-point interpolated AP averaged over IoU thresholds 0.50:0.05:0.95, alongside AP50, AP75 and size-based APS, APM and APL. We also report per-class AP50, inference speed (FPS) and an error breakdown of misses and false positives. Model selection uses the validation split, and final numbers are reported on the held-out test split.

V. RESULTS

Table I summarises test performance. Our improved Faster R-CNN (ResNet-50-FPN v2 with RFS and underwater augmentation) is the strongest of our models, raising Test AP50 from 18.9% to 21.4% (+2.5) and Test AP from 10.7% to 12.1% (+1.4) over the ResNet-50 v1 baseline. Gains are spread across the board, with AP75 rising from 10.5% to 12.7% and small-object APS rising from 4.0% to 6.2%. Interestingly, validation AP50 only moved from 22.7% to 23.2%, so most of the improvement shows up on the held-out test videos, which suggests the new recipe mainly helps the model generalise to unseen footage.

TABLE I. TEST RESULTS ON TRASHCAN-MATERIAL 1.0

Model	FPS	AP	AP50	APS	APL
MobileNetV3-L-FPN (W7)	127	8.8	17.7	1.6	12.6
R50-FPN v1 (W7)	52	10.7	18.9	4.0	15.2
RetinaNet R50-FPN v2	68	7.2	11.6	6.3	10.3
R50-FPN v2 + RFS + Aug	**49**	**12.1**	**21.4**	**6.2**	**16.2**
X101-FPN [1]*	9	29.1	51.2	28.2	40.0

[a] *Reported by [1] on its own validation split with a 104M parameter model; not directly comparable.*

Per class, the best results are animal_eel (59.7% AP50), rov (57.4%) and animal_crab (51.9%). RFS lifts tail classes across all four families, though AP50 still correlates

with log class frequency ($r = 0.44$). The hardest classes remain trash_fishing_gear (2.6%), trash_rubber (0.3%) and the catch-all trash_etc (20.3%). Qualitatively, the model reliably separates debris such as plastic bags, bottles and cans from nearby animals and ROV parts, and still gives confident (>0.80) boxes in murky blue-green frames.

The error analysis shows modest but consistent gains: large-object misses drop from 64.9% to 63.1%, dark-frame misses from 65.3% to 63.3%, and the localisation false-positive rate from 11.3% to 10.4%. Small objects (under 32^2 px) remain the main weakness, with an 85.6% miss rate.

Comparing architectures, Faster R-CNN v2 (21.4% Test AP50) far outperforms RetinaNet (11.6%), despite RetinaNet's focal loss being designed for imbalance. With only 1.7 objects per frame, the RPN's proposal filtering followed by RoIAlign appears to handle sparse underwater scenes better than a dense anchor grid, where background dominates. MobileNetV3 offers a useful trade-off for embedded use, reaching 17.7% Test AP50 at 127 FPS.

In terms of real-world deployment, our models are not yet ready: missing around two in three objects is too high for autonomous clean-up, though they could support semi-automated surveys with a human in the loop. There is also a clear gap to the ResNeXt-101 baseline of [1] (51.2% AP50). That model has 2.5 times the parameters of ours and runs at roughly 9 FPS, and is evaluated on a different split, so the numbers are not directly comparable. Our improved model runs over five times faster at 49 FPS, and our video-disjoint split gives a stricter, more honest estimate of performance on new dives.

VI. CONCLUSION

Through this project, we benchmarked Faster R-CNN and RetinaNet detectors on TrashCan-Material 1.0 under a video-disjoint protocol, and showed that improvements targeted at the dataset's specific problems, namely RFS for class imbalance and colour/contrast augmentation for underwater degradation, raise Test AP50 from 18.9% to 21.4% at 49 FPS. Our exploratory analysis also identified frame-level leakage as a real risk in this dataset, which matters for any fair comparison of future models. A key strength of our solution is that it remains fast enough for onboard use, and our comparison shows two-stage detectors are the better fit for sparse underwater scenes.

The main limitations are poor small-object detection (85.6% miss rate), confusion within trash families and the catch-all trash_etc class, and a large remaining gap to heavier models. We also only evaluated bounding boxes on the Material version. Given more time, we would compare box and instance-mask supervision with Mask R-CNN on both the Material and Instance versions, test multi-scale test-time augmentation and higher-resolution inputs for small objects, and benchmark single-stage YOLO models, which [2] found to perform best.

VII. ACKNOWLEDGEMENTS

Debating glazing tutor

VIII. REFERENCES

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